

Advanced (Habanero)

- Q1.** If $f(x) = x^2$ and $g(x) = 2x$, find $f(g(2))$.
(A) 8
(B) 16
(C) 32
(D) 64
(E) 128
- Q2.** Given $f(x) = \frac{1}{x}$ and $g(x) = x - 1$, determine the domain of $f(g(x))$.
(A) all real numbers
(B) all real numbers except $x = 1$
(C) all real numbers except $x = 0$
(D) all real numbers except $x = -1$
(E) $x > 0$
- Q3.** If $f(x) = |x|$ and $g(x) = -x$, find $f(g(-2))$.
(A) 2
(B) -2
(C) 0
(D) 1
(E) -1
- Q4.** For the functions $f(x) = x^2$ and $g(x) = \sqrt{x}$, find $f(g(4))$.
(A) 2
(B) 4
(C) 8
(D) 16
(E) 32
- Q5.** Given $f(x) = x + 2$ and $g(x) = 2x - 1$, find $g(f(2))$.
(A) 5
(B) 7
(C) 9
(D) 11
(E) 13
- Q6.** If $f(x) = \frac{1}{x+1}$ and $g(x) = x - 2$, determine the domain of $g(f(x))$.
(A) all real numbers except $x = 1$
(B) all real numbers except $x = 0$
(C) all real numbers except $x = -1$
(D) all real numbers except $x = -2$
(E) $x > -1$
- Q7.** For $f(x) = x^3$ and $g(x) = x - 1$, find $f(g(1))$.
(A) 0
(B) 1
(C) 8
(D) 27
(E) 64
- Q8.** Given $f(x) = \sqrt{x}$ and $g(x) = x^2$, find $g(f(4))$.
(A) 2
(B) 4
(C) 16
(D) 32
(E) 64

Open Ended Questions (FRQ)

- Q9.** A water tank can be filled at a rate of $f(t) = 2t$ gallons per minute. However, due to a leak, water is also being lost at a rate of $g(t) = t^2$ gallons per minute. Determine the net rate at which the tank is being filled after $t = 2$ minutes.
- Q10.** Suppose $f(x) = x^2 + 1$ represents the cost of producing x units of a product, and $g(x) = 3x$ represents the revenue from selling x units. Find the profit $P(x)$ from selling x units and interpret $P(10)$.

Chef's Tips

- Emphasize the importance of understanding function composition and its applications.
- Encourage students to carefully consider domain restrictions for composite functions.
- Remind students to interpret results in the context of real-world problems.